1 **1942**

A theropod skull was found in Hell Creek Formation rocks from Montana. This formation is known for producing Tyrannosaurus rex specimens, and while the skull closely resembles a T. rex, it is only about half the length.

2 **1965**

Some small tyrannosaur skeletons from Mongolia, once thought to be separate species, are young Tarbosaurus.

3 **1988**

The 1942 skull has 15 teeth in one side of its upper jaw. Adult T. rex have 11 or 12.

4 **1999**

The joints between the bones of the 1942 skull are open, not fused.

5 **1999**

In Gorgosaurus, young animals have more teeth than adults: tooth number falls as the animal grows.

6 **2001**

A second small tyrannosaur from Hell Creek (Jane), is found with a more complete skeleton. Jane's skull is lower and narrower than an adult T. rex's.

7 **2003**

In Albertosaurus, Gorgosaurus and Daspletosaurus, young animals have low, narrow skulls and adults have deep, heavy skulls.

8 **2011**

A juvenile Tarbosaurus has about the same number of teeth as adult Tarbosaurus.

9 **2013**

Bloody Mary, a third small tyrannosaur from Hell Creek, has hand proportions and a wishbone (furcula) shape that differ from T. rex.

10 **2020**

Bone sections from Jane show growth rings and bone tissue typical of an animal that was still growing.

11 **Oct 2025**

The outermost growth rings in Bloody Mary's bone are tightly packed, a pattern seen when growth has nearly stopped.

12 **Oct 2025**

Growth rings put Bloody Mary at 17 to 22 years old at death, and Jane at 8 to 14.

13 **Oct 2025**

Bloody Mary has 16 teeth in the left side of its upper jaw and 17 in the right. Adult T. rex have 11 or 12.

14 **Oct 2025**

Bloody Mary's hand bones are larger than an adult T. rex. The 20' long Bloody Mary has a first finger almost twice the length of a 42' long T. rex.

15 **Oct 2025**

Bloody Mary has fewer tail vertebrae than T. rex.

16 **Oct 2025**

Bloody Mary has several air-sac openings (pleurocoels) in its second neck vertebra and in its tail vertebrae. Jane has one in the second neck vertebra and none in the tail.

17 **Dec 2025** Bone sections from the throat bones (hyoids) of the 1942 skull show it was a mature animal.	**18** **Jan 2026** Growth patterns in Jane’s bone do not match the growth curve built from T. rex hindlimbs of many sizes.